

Murtaza Khalid | Software Engineer

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Software Engineer with 3 years of experience in building backend systems at national scale, including Pakistan's RTGS interbank settlement platform and RAAST instant payment infrastructure — two of the country's most critical financial systems. Deeply skilled in Java, Spring Boot, and microservices, with a strong track record in building resilient RESTful APIs, ISO 20022/SWIFT-compliant messaging pipelines, and high-throughput data systems. Hands-on with RabbitMQ, Redis, Docker, Liquibase, Keycloak, AWS (EC2, RDS, S3, SQS), and Apache Kafka, spanning database management, performance optimization, cloud infrastructure, event streaming, and secure authentication flows — consistently delivering solutions that are reliable, compliant, and built to scale.

TECHNICAL SKILLS

Java | Spring | Spring MVC | Spring Boot | Spring Security | Hibernate | Microservices | REST APIs | RabbitMQ | Redis | SQL Server | MySQL | Oracle DB | Maven | Gradle | Docker | Git | GitLab CI | GitHub Actions | JUnit & Mockito | Jira | Postman | Linux | Liquibase | Keycloak | Multimodule Projects | AWS (EC2,RDS,S3, SQS) | Apache Kafka | DBeaver | WinSCP | ISO 20022 | SWIFT MT/MX | SBP Compliance

PROFESSIONAL EXPERIENCE

Software Engineer at Paysys-Labs

July 2024 - Present

Engineered **backend systems** for two of **Pakistan's** most critical national payment platforms **RTGS (Real-Time Gross Settlement)** for high-value interbank settlements and **RAAST**, the country's instant retail payment infrastructure serving the entire Pakistani banking network.

• RTGS– Real-Time Gross Settlement System:

Owned end-to-end backend development of Pakistan's national RTGS platform, handling high-value interbank settlements, customer fund transfers, and government securities transactions

- **Engineered** the migration of **RTGS** from legacy **SWIFT MT** message workflows to **ISO 20022 MX**-compliant messaging, aligning with **State Bank of Pakistan's** modernization mandate..
- **Designed and implemented** services to parse MT messages, extract structured financial data, and generate MX messages for integration with SBP's payment infrastructure.
- Ensured compliance with international financial messaging standards (**ISO 20022, SWIFT**) and maintained operational reliability across transaction processing pipelines..
- **Authored** technical documentation to support regulatory compliance requirements and team knowledge transfer.
- **Conducted** performance testing and profiling to ensure system stability and reliability under **real-time** high-value transaction loads.
- The system uses **AWS EC2, leveraging S3 buckets** for secure document and message storage, and **Apache Kafka** and **SQS** as the backbone for inter-service communication and **event-driven** message flow.

• RAAST – Instant Payment System:

Contributed to backend development of **RAAST — Pakistan's national instant payment** rail supporting real-time, fee-free transactions for retail banking

- Developed backend features for **Person-to-Person (P2P)** transaction flows including **RAAST** ID creation, linking, and delinking.
- Worked on **Person-to-Merchant (P2M)** payment APIs enabling seamless merchant transactions across the **RAAST** ecosystem.
- Integrated bill payment APIs for major telecom providers including **Jazz** and **Ufone**, enabling instant utility and telecom payments through the **RAAST** network.
- Implemented **SAF (Store and Forward)** protocol to ensure transaction reliability and delivery guarantees during network disruptions.
- Worked on **hugo digital bank** onboarding backend workflows including identity verification integrations with **NADRA BVS** and **CNIC** verification systems, ensuring regulatory-compliant customer onboarding.
- Ensured secure communication with **RAAST** infrastructure using standardized APIs and protocols in compliance with **SBP** scheme rules.

Trainee Software Engineer at Stepway-Softwares

Aug 2023 - June 2024

Developed the web applications for logistics, e-commerce, and data visualization, with a primary focus on a logistics management system for a **Gulf-based** courier company.

- Developed the backend for the Logistics System (**SMSA**), building both admin and user panels to manage parcel tracking, automate **bulk data processing**, and generate pseudo-customer records; implemented role-based access control and performance-focused UI features like pagination to enhance usability.
- Deployed and managed the application on an **on-premises server**, ensuring reliability, smooth releases, and system availability.
- Designed and implemented a high-volume data pipeline using **Spring Batch** to process bulk Excel files, inserting thousands of records into the database efficiently; integrated automated email notifications to inform customers with their order tracking and shipment IDs.

EDUCATION

Mohammad Ali Jinnah University ,Karachi ,Pakistan

Sep 2020 - July 2024

- Bachelours Of Science In Computer Science (BSCS) CGPA 3.5.

PERSONAL PROJECTS

- Lead Generation –App -Flutter, Spring Boot and Angular.
- Point Of Sale – Java Swing and java Fx.
- Bank Management System – Core Java.
- Todo List – Android.

IMPORTANT-LINKS

- <https://github.com/murtazakhalid77>
- <https://www.linkedin.com/in/murtazakhalid77/>
- <https://leetcode.com/u/murtazakhalid77/>